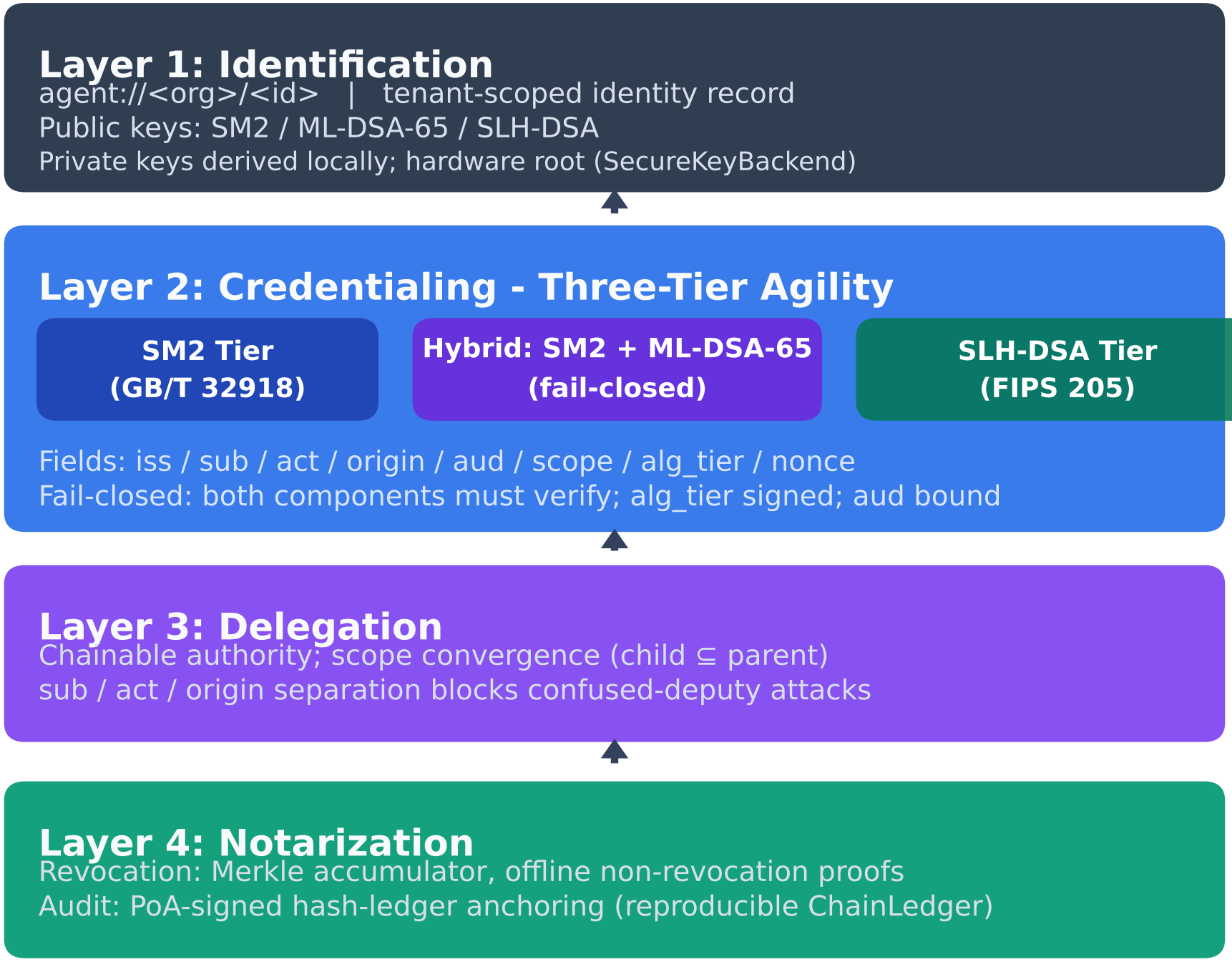
Layer 1: Identification
agent://<org>/<id> | tenant-scoped identity record
Public keys: SM2 / ML-DSA-65 / SLH-DSA
Private keys derived locally; hardware root (SecureKeyBackend)
Layer 2: Credentialing - Three-Tier Agility
SM2 Tier (GB/T 32918)
Hybrid: SM2 + ML-DSA-65 (fail-closed)
SLH-DSA Tier (FIPS 205)
Fields: iss / sub / act / origin / aud / scope / alg_tier / nonce
Fail-closed: both components must verify; alg_tier signed; aud bound
Layer 3: Delegation
Chainable authority; scope convergence (child ⊆ parent)
sub / act / origin separation blocks confused-deputy attacks
Layer 4: Notarization
Revocation: Merkle accumulator, offline non-revocation proofs
Audit: PoA-signed hash-ledger anchoring (reproducible ChainLedger)